

While Loop, For loop and Function

1. Number Guessing Game(While loop)

This program creates a simple number guessing game using a while loop. First, the random module is imported and the randint() function generates a random number between 1 and 10, which is stored as the secret number. The user is given three attempts to guess the number. A while loop repeats the guessing process, while if, elif, and else statements compare the user's guess with the secret number. If the user enters a number outside the range of 1 to 10, an invalid input message is displayed, and the continue statement skips the current iteration. If the guess is greater than the secret number, the program displays "Number is too high", and if it is smaller, it displays "Number is too low." When the user guesses the correct number, the program displays "Number is correct" and the break statement stops the loop. The number of attempts is reduced after every valid incorrect guess. This program demonstrates random numbers, while loops, conditional statements, break, continue, and user input.

2. Multiplication Table (For loop)

This program generates the multiplication table of a number entered by the user. First, the program uses input() to get a number from the user and int() converts the input into an integer. A for loop with range(1, 11) is then used to repeat the multiplication from 1 to 10. During each iteration, the value of i represents the multiplier, and num * i calculates the result. The program displays each multiplication step in a clear format. This program helps in understanding the use of user input, for loops, range(), variables, and arithmetic operators in Python.

3. BMI Calculator Using a Function

This program calculates the Body Mass Index (BMI) using a user-defined function called calculate_bmi. The function accepts two parameters, weight and height, and calculates BMI using the formula $BMI = Weight / Height^2$. The calculated value is returned using the return statement. The program then prompts the user to enter their weight in kilograms and height in meters. The float() function converts the input values into decimal numbers. Finally, the calculate_bmi() function is called by passing the user's weight and height as arguments, and the calculated BMI is displayed. This program demonstrates user-defined functions, parameters, arguments, return statements, user input, type conversion, and arithmetic operations.

Outputs:

```
... Enter the number:7
    Number is too low
    Enter the number:9
    Number is too high
    Enter the number:11
    Invalid input range,Give input range between 1 to 10
    Enter the number:6
    Number is too low
    Enter the number:8
    Number is correct
```

```
... Enter the number:5
    5 x 1 = 5
    5 x 2 = 10
    5 x 3 = 15
    5 x 4 = 20
    5 x 5 = 25
    5 x 6 = 30
    5 x 7 = 35
    5 x 8 = 40
    5 x 9 = 45
    5 x 10 = 50
```

```
Enter your weight in kg: 74
Enter your height in meters: 1.75
Your BMI is: 24.163265306122447
```